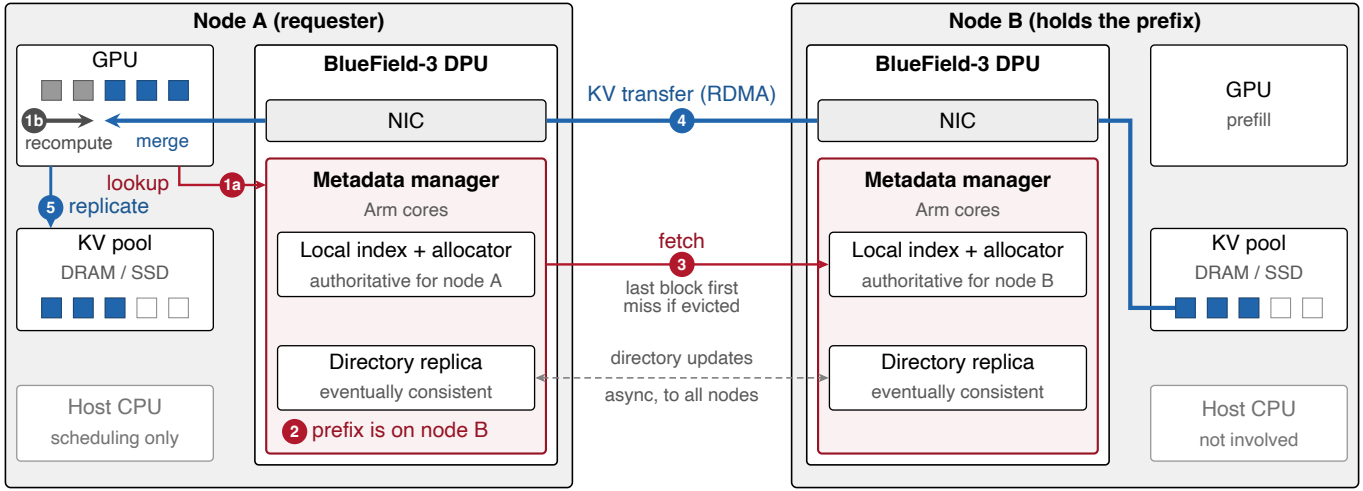
Node A (requester)
GPU
1b
recompute
merge
lookup
1a
5
replicate
KV pool
DRAM / SSD
Host CPU
scheduling only
BlueField-3 DPU
NIC
Metadata manager
Arm cores
Local index + allocator
authoritative for node A
Directory replica
eventually consistent
2
prefix is on node B
KV transfer (RDMA)
4
fetch
3
last block first
miss if evicted
directory updates
async, to all nodes
Node B (holds the prefix)
BlueField-3 DPU
NIC
Metadata manager
Arm cores
Local index + allocator
authoritative for node B
Directory replica
eventually consistent
GPU
prefill
KV pool
DRAM / SSD
Host CPU
not involved